Appendix: Connectivity-Preserving Exact Search for Non-Hereditary Cohesive Subgraphs: Computing Maximum Flexi-Cliques

A Hardness proof of maximum Flexi−clique problem

A.1 NP-hardness

LEMMA 1. *The Flexi−clique decision problem is* NP*-hard.*

PROOF. We prove NP-hardness by reduction from the well-known *k-clique* problem, which is NP-complete in its decision form. Given an instance (G', k) of the clique decision problem, we construct an equivalent instance (G, τ, k) of the Flexi−clique decision problem as follows.

We set $G = G'$, meaning that the node and edge sets remain unchanged. We then choose $\tau \in [0, 1)$ sufficiently close to 1 so that for any subset of size x, the degree threshold satisfies $\lfloor x^\tau \rfloor = x - 1$. Intuitively, this setting makes the Flexi−clique condition equivalent to the clique constraint. Formally, it suffices to select τ satisfying

$$\frac{\ln(|V| - 1)}{\ln(|V|)} < \tau < 1,$$

since for any $x < x'$, the ratio $\frac{\ln(x-1)}{\ln(x)} < \frac{\ln(x'-1)}{\ln(x')}$. The parameter k remains the same.

If G' contains a clique H of size at least k, then H also forms a valid Flexi−clique in G, because each node has degree $|H| - 1$ within H. Conversely, if G contains a Flexi−clique H of size at least k, the degree constraint forces H to be a clique, implying that G' contains a clique of size at least k. Since the transformation runs in polynomial time, the Flexi−clique decision problem is NP-hard. □

LEMMA 2. *The Maximum Flexi−clique problem is* NP*-hard.*

PROOF. We prove NP-hardness by reducing from the Flexi−clique decision problem (shown to be NP-hard in Lemma 1).

Suppose we have a polynomial-time algorithm $\mathcal{A}$ that finds the maximum Flexi−clique in any instance (G, τ). Given an instance (G, τ, k) of the k-Flexi−clique problem, we:

(1) Use $\mathcal{A}$ to compute the largest Flexi−clique $H_{\max}$ and its size $|H_{\max}|$.

(2) If $|H_{\max}| \geq k$, return yes; otherwise, return no.

Since it solves the NP-hard k-Flexi−clique problem in polynomial time, the Maximum Flexi−clique problem must also be NP-hard. □

A.2 Constant factor inapproximability

THEOREM 1. *The Maximum Flexi−clique problem is constant-factor inapproximable unless* P = NP.

PROOF. We prove the claim via a gap-preserving reduction from the gap version of the CLIQUE problem. Specifically, given a graph $G = (V, E)$, an integer K, and a constant $c > 1$, the task is to distinguish between the following two cases, under the promise that one of them holds:

- **YES**: G contains a clique of size at least K;
- **NO**: every clique in G has size at most K/c.

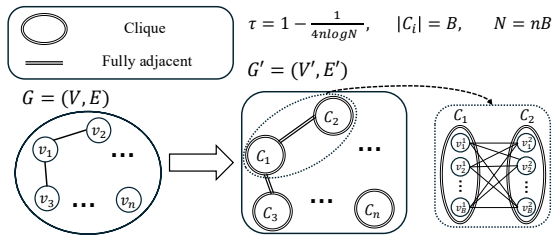

Fig. 1. Reduction process

It is NP-hard to distinguish between the above two cases [1].

Given a graph $G = (V, E)$ with parameters K and $c > 1$, under the promise that G contains a clique of size at least K or that every clique in G has size at most K/c, we construct a graph $G' = (V', E')$ that preserves the promised gap while amplifying differences in clique size. Specifically, for each node $v \in V$, we create a clique C_v consisting of exactly B nodes, where B is a sufficiently large polynomial in n. The node set of G' is defined as $V' = \bigcup_{v \in V} C_v$, and we denote $|V'|$ by N. For any two distinct nodes $u, v \in V$, the cliques C_u and C_v are made fully adjacent if $(u, v) \in E$, and completely non-adjacent otherwise. Next, we set the parameter τ as

$$\tau = 1 - \frac{1}{4n \log N},$$

which will be used to control the gap between s and s^τ in the analysis below. The graph construction process for the gap-preserving reduction is illustrated in Figure 1. For the constructed graph G', we denote by $\mathrm{OPT}(G')$ the maximum size of a Flexi−clique.

YES Case. Assume that G contains a clique $Q \subseteq V$ with $|Q| = K$, and let $S = \bigcup_{v \in Q} C_v \subseteq V'$. Clearly, the set S has size KB, and the induced subgraph $G'[S]$ is connected since all cliques corresponding to nodes in Q are adjacent. Consider any node $x \in S$, since C_v is a clique, x has $B - 1$ neighbours inside $G'[C_v]$. Moreover, from the other $K - 1$ cliques in S, the node x has in total $(K - 1)B$ neighbours. Therefore,

$$\deg_{G'[S]}(x) \geq (B - 1) + (K - 1)B = KB - 1. \tag{1}$$

As $\tau \in [0, 1)$, $KB - 1 \geq \lfloor (KB)^\tau \rfloor$ for the chosen value of B. Hence $\deg_{G'[S]}(x) \geq \lfloor |S|^\tau \rfloor$ for all $x \in S$, and thus S is a valid Flexi−clique with $|S| = KB$, implying

$$\mathrm{OPT}(G') \geq KB. \tag{2}$$

NO Case. Now assume that every clique in G has size at most K/c. Let $S \subseteq V'$ be an arbitrary Flexi−clique in G'. For each $v \in V$, define $a_v = |S \cap C_v|$, and let

$$U = \{v \in V : a_v > 0\}. \tag{3}$$

Denote

$$s = |S| = \sum_{v \in U} a_v. \tag{4}$$

We first bound the total number of non-neighbours that a node in a Flexi−clique can have. Recall that $\tau = 1 - \frac{1}{4n \log N}$ and that $s \le nB = N$. Thus,

$$s - s^{\tau} \le \frac{s}{4n} \le \frac{B}{4}. \tag{5}$$

Fix any $u \in U$ and consider an arbitrary node $x \in S \cap C_u$. By construction of G', the degree of x in the induced subgraph $G'[S]$ is

$$\deg_{G'[S]}(x) = (a_u - 1) + \sum_{\substack{v \in U: \\ (u,v) \in E}} a_v. \tag{6}$$

Therefore, the number of nodes in S that are non-adjacent to x is

$$(s-1) - \deg_{G'[S]}(x) = \sum_{\substack{v \in U: \\ (u,v) \notin E}} a_v. \tag{7}$$

Since S is a Flexi−clique, we have $\deg_{G'[S]}(x) \ge \lfloor s^{\tau} \rfloor$, and hence

$$\sum_{\substack{v \in U: \\ (u,v) \notin E}} a_v \le (s-1) - \lfloor s^{\tau} \rfloor \le s - s^{\tau} \le \frac{B}{4}, \tag{8}$$

where the last inequality follows from (5). Now let C be a maximum clique in the induced graph $G[U]$. By the NO-case assumption, we have $|C| \le K/c$. By maximality of C, for every $v \in U \setminus C$ there exists a node $u \in C$ such that $(u, v) \notin E$, and we assign each $v \in U \setminus C$ to one such u. For any fixed $u \in C$, it follows from (8) that

$$\sum_{v \in U \setminus C} a_v \le \sum_{\substack{v \in U: \\ (u,v) \notin E}} a_v \le \frac{B}{4}. \tag{9}$$

Summing over all $u \in C$, we obtain

$$\sum_{v \in U \setminus C} a_v \le |C| \cdot \frac{B}{4}. \tag{10}$$

Since each block C_v has size exactly B, we also have

$$\sum_{u \in C} a_u \le |C| B. \tag{11}$$

Combining this with (10), we conclude that

$$\begin{aligned} |S| &= \sum_{u \in C} a_u + \sum_{v \in U \setminus C} a_v \\ &\le |C|B + |C| \cdot \frac{B}{4} = \frac{5}{4}|C|B \le \frac{5}{4} \cdot \frac{K}{c} B. \end{aligned} \tag{12}$$

Therefore,

$$\mathrm{OPT}(G') \le \frac{5}{4} \cdot \frac{K}{c} B. \tag{13}$$

In conclusion, combining (2) and (13) yields a constant gap between **YES** and **NO** instances. Therefore, the Maximum Flexi−clique problem does not admit any constant-factor approximation unless P = NP. □

References

[1] Uriel Feige, Shafi Goldwasser, László Lovász, Shmuel Safra, and Mario Szegedy. 1991. Approximating clique is almost NP-complete. In *[1991] Proceedings 32nd Annual Symposium of Foundations of Computer Science*. IEEE Computer Society, 2–12.